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Efficient Pattern-Based Analysis of Network Flows in IoT Systems

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TwinODIS

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FOUNDATION FOR RESEARCH AND TECHNOLOGY - HELLAS

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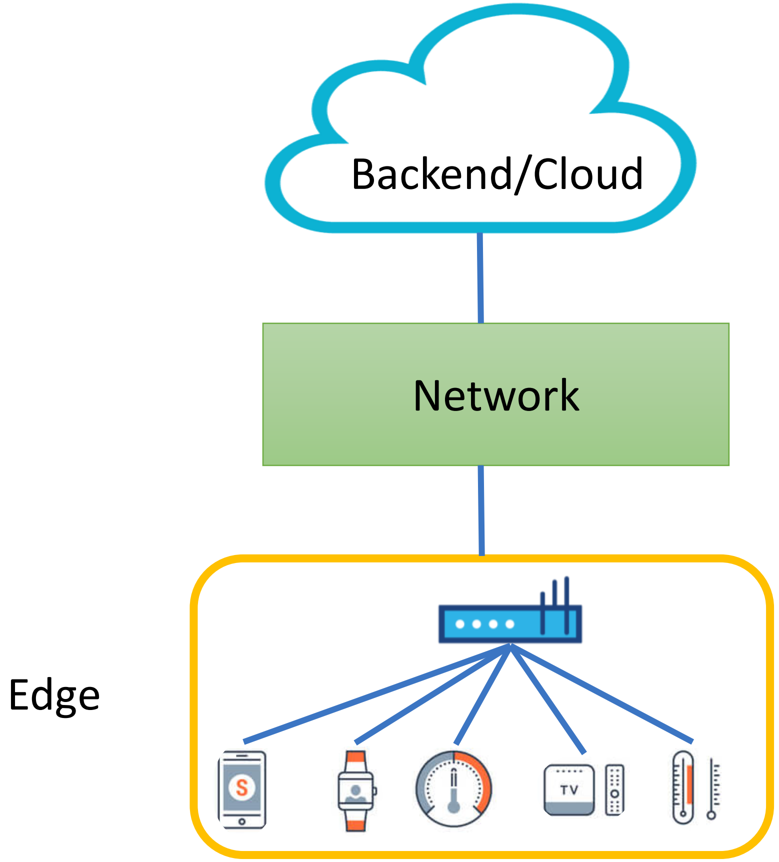
Internet-of-Things (IoT) has advanced the concept of Internet networking



- Smart Transportation & Mobility
- Smart Energy Grids
- Healthcare & e-Health Monitoring
- Smart Industry / Industry 4.0
- Disaster Management & Environmental Monitoring

- Smart objects, IoT applications and platforms vulnerable to security attacks
- Massive personal data generation
- Preserving security is challenging
  - Analyzing vulnerabilities
  - Selecting appropriate controls
  - End-to-end security preservation under dynamic conditions

Network Flow

IoT-device flows

Timestamp	Device ID	Protocol	Src->Dst	Pkts	Bytes	Δt	Port
12:00	CAM-01	TCP/443	10->52	14	19K	220	443
12:00	CAM-01	TCP/443	52->10	11	3K	60	443
12:00	BULB-05	UDP/5683	11->53	2	120	15	5683
12:01	TEMP-07	MQTT/1883	12->54	4	512	110	1883

CAM-01 snapshot


```
{
 "device_id":  "CAM-01",
 "proto":     6,              // TCP
 "dst_port":   443,
 "seq_len":    14,
 "pkt_sizes":  [66,-66,52,1368,-60,1368,-60,1368,-60,1368,-60,1368,-60,52],
 "delta_ms":   [30,2,102,33,7,31,8,31,9,31,8,32,8],
 "start_time": "2025-06-16T12:00:00.000Z"
}
```


TEMP-07 snapshot


```
{
 "device_id":  "TEMP-07",
 "proto":     6,              // TCP
 "dst_port":   1883,            // MQTT
 "seq_len":    4,
 "pkt_sizes":  [150, -62, 240, -60],
 "delta_ms":   [40, 40, 30],
 "start_time": "2025-06-16T12:01:00.000Z"
}
```


SMART CITIES

Network Flow

Signature-based pattern matching

- » Signature: known temporal or structural pattern observed in network traffic
  - › Can be based on packet size, timing, direction, or protocol
- » Widely used in:
  - › Intrusion detection systems e.g. Snort, Suricata
  - › Network-based analytics e.g. Cisco Stealthwatch
- » Patterns often extracted from:
  - › Threat intelligence feeds
  - › Forensic analysis of past incidents

Packets sent	0,	0,	50,	0,	0,	50,	0,	0,	50,	0,	0,	50,	0,	0,	50,	0,	0,	50
10-sec window	0	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170

Network Flow

- Variable-length flow patterns
- High data velocity and volume
- Need for low-latency, adaptive indexing

Proposed Method

- Scalable indexing for variable-length subsequences
- Single index structure for multiple query lengths
- Works with Z-normalized & non-Z-normalized data
- Supports both Euclidean & DTW queries

The VLDB Journal (2020) 29:1449–1474
https://doi.org/10.1007/s00778-020-00619-4

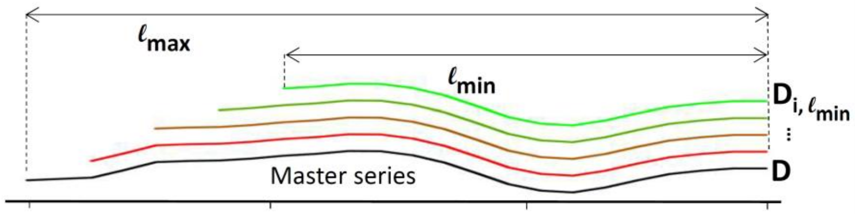
REGULAR PAPER

Scalable data series subsequence matching with ULISSE

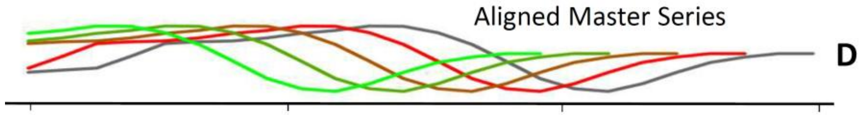
Michele Linardi[1] · Themis Palpanas[1]

Proposed Method

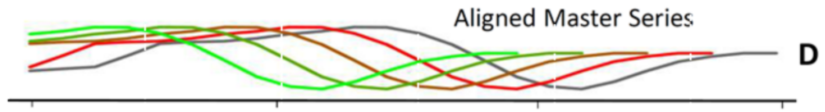
1 Pick master series

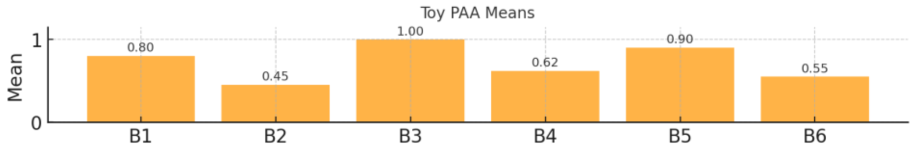


2 Zero-align series

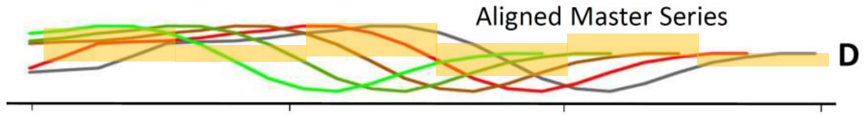


3 Piecewise aggregate Approximation (PAA) summarize

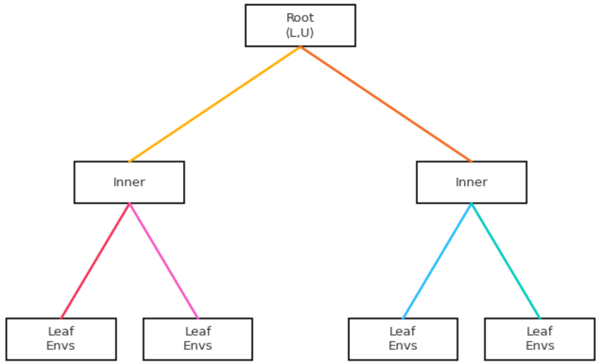




4 Build the envelope (uENV)



5 Insert into tree



Proposed Method

Multi-feature fusion

- ✓ Concatenate packet size, inter-arrival, protocol ID vectors
- ✓ Weighted Euclidean across dimensions

Real-time incremental indexing

- ✓ Implementing online index updates as new traffic arrives
- ✓ Avoid costly reindexing while ensuring responsiveness

Contextual partitioning

- ✓ Separate sub-indexes by device class (e.g., sensors, cameras, etc.)

Template-based querying

- ✓ Incorporate known attack and anomaly templates for rapid similarity queries using k-NN over the index

Workflow

Raw IoT Traffic → **Sequences Extraction** → **ULISSE Subsequence Index** → **Similarity Queries** → **Pattern Detection & Alerting**

Raw IoT Traffic

Stream ingest
(pcap → flows)

Sequences Extraction

ULISSE Subsequence Index

Dual indexing
-> In-memory hash (recent)
-> ULISSE tree (historical)

Similarity Queries

Query:
-> Incoming window
-> Templates

Pattern Detection & Alerting

Alert/logs/visualize

Evaluation Protocol

- ✓ TON IoT
- ✓ CICIDS
- ✓ UNSW HomeNet

- ✓ Detection Accuracy
- ✓ Query Latency
- ✓ Indexing Throughput
- ✓ Scalability & Efficiency

- ✓ Suricata (signature-based)
- ✓ SAX-based methods
- ✓ DTW + Sliding Window

Expected Benefits

- High detection accuracy (multi-feature similarity)
- Support for evolving patterns
- Low query latency, real-time alerts
- Efficient memory footprint
- Scalable to large IoT deployments

Broader Impact

Deployable on edge gateways or cloud platforms

Adaptive learning integration (next phase)

Enhances both security & activity recognition

THANKS FOR WATCHING!

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